MIL-W-80103C
24 July 1986
SUPERSEDING
MIL-W-80103B
4 November 1974

MILITARY SPECIFICATION

WELDING EQUIPMENT, ARC, PIPE OR TUBE, AUTOMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers automatic welding equipment for circumferential welding of pipes or tubes in both the horizontal and vertical positions, using the gas metal arc (MIG) or gas tungsten arc (TIG) processes with equipment and accessories specified herein.

1.2 Classification. The welding equipment shall be of the following types. The type to be furnished shall be as specified (see 6.2.1).

Type I - Exposed, orbital welding head (TIG or MIG) on a clamp-on type stationary unit. Pipe or tube is stationary.

Type II - Enclosed, orbital welding head (TIG) on a clamp-on type stationary unit. Pipe or tube is stationary.

Type III - Exposed, stationary welding head (TIG or MIG) mounted on a torch arm assembly. Pipe or tube is revolved by unit.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-278 (SH) - Fabrication, Welding and Inspection Machinery, Piping and Pressure Vessels for Ships of the United States Navy.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the nongovernment documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA EW 1 - Electric Arc-Welding Apparatus.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

AGMA 360.02 - Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specifications sheets or military standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding equipment shall be new and of the manufacturer's current models. The pipe welding equipment specified herein shall be designed to weld standard pipe or tubing as described in 3.4. The equipment shall be either an orbital or stationary type. Types I and II shall be designed to weld stationary pipe or tubing. Type III shall be designed to weld pipe or tubing which is rotated. All types shall have a power supply, master control, welding heads, and travel or positioning devices as described herein. The welding equipment shall provide control of welding operations, including but not limited to travel speed, welding current, voltage, oscillation, and dwell. The controls shall be easily accessible to the operator. All parts subject to wear shall be easily accessible for adjustment, replacement and repair.

3.2.1 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or the welding equipment.

3.2.2 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding equipment. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.3 Materials. Materials shall be as specified herein. Materials not specified herein shall be selected by the manufacturer to meet the requirements of this specification. Contact between dissimilar metals, especially brass, copper, or steel in contact with aluminum or magnesium shall be avoided whenever possible. Materials shall be free of all defects and imperfections that might affect the serviceability and appearance of the finished product.

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3.2.3.1 Reclaimed materials. The welding equipment may contain reclaimed materials provided such materials will not jeopardize the welding equipment's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the welding equipment.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding equipment that present safety hazards. The safety devices shall not interfere with the operation of the welding equipment. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All welding equipment parts, components, mechanisms, and assemblies furnished on the welding equipment whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR that are applicable to the welding equipment itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.5 Mercury restriction. The welding equipment shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding equipment.

3.2.7 Environmental protection. The welding equipment shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the welding equipment shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.8 Energy efficiency. The welding equipment and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.3 Construction. The welding equipment shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the welding equipment to its maximum rating and capacity without permanent distortion or damage.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

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3.3.2 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding equipment shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the welding equipment and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears and pinions of spindle and axis drive trains shall be desinged and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.4 Performance. The welding equipment shall be capable of repeatedly making sound, acceptable pipe or tube welds in both the horizontal and vertical positions. Welding parameters such as travel speed, wire feed, welding current, voltage, oscillation frequency and dwell time shall be controllable from a location easily accessible to the operator. The welding equipment shall produce welded joints which meet the test requirements of MIL-STD 278 (SH) on class P-1 piping and radiographic acceptance standards for welded P-1 piping. The welding equipment shall be rated at 100 percent duty cycle.

3.5 Components. The welding equipment shall consist of, but not be limited to the following components: power supply, master control, travel or positioning devices, welding heads, and cables and hoses.

3.5.1 Power supply. The welding power supply shall be capable of supplying either or both constant current and constant voltage. The power supply shall be in accordance with the basic requirements of applicable military specifications as specified (see 6.2.1).

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3.5.2 Master control. The master control shall be a console type wherein voltage, down slope, current, start, stop, carriage speed, wire feed rate, arc gap, pulse, amplitude, oscillation and dwell shall be sequence controlled. The master control and power supply may be in the same enclosure. When specified (see 6.2.1), a pendant type control or controls mounted on the unit attached to the pipe shall be provided to control start and stop of the welding. When gas metal arc process or a wire feed gas tungsten arc process is used, the following additional controls shall be provided but not be limited to: an inch push button switch to properly position the wire to the pipe joint and weld set-up switch to start the welding with an antistick time delay. The master control shall automatically control the voltage, current, and slope. When gas tungsten arc process without wire feed is used, automatic controls for programming of high frequency arc starting or touch/retract start, welding current pulsation, travel speed and shielding gas for the arc and for backup shall be provided in addition to above controls. All controls shall be provided with marked scales to designate a setting for repeat performances. Controls shall be free from binding and constructed so that a fixed setting shall remain constant as set.

3.5.3 Travel device. The traveling device for the three types of pipe welding equipment shall be as specified in 3.5.3.1, 3.5.3.2, and 3.5.3.3.

3.5.3.1 Type I travel device. The travel device for type I pipe welding equipment shall be capable of operating on pipe with diameters ranging from not less than 3 inches (CM) to not more than 8 inches and with any standard wall thickness. The travel device shall consist of a welding head, a clamping device which will firmly secure the travel device to the pipe, and a rotational drive unit with a guiding device which will provide constant rate and accurate direction of travel. Installation and removal of the equipment shall be free from the requirement of access to pipe ends.

3.5.3.2 Type II travel device. The travel device for type II pipe welding equipment shall be compact and lightweight. The travel device shall not exceed 6 inches radial distance around the outside diameter of the pipe on three quadrants and shall not exceed 9 inches in the longitudinal direction of the pipe. The traveling device shall have clamping devices to hold the unit firmly to the pipe. The unit shall orbit an enclosed gas tungsten arc head 360 degrees around the pipe. Provisions for oscillation, dwell and controlling the arc shall be provided. Clamping devices for pipe size from 1 inch to 6 inches shall be supplied with the equipment. Controls for the traveling unit shall be on the master control.

3.5.3.3 Type III travel device. The type III travel device shall consist of a means to rotate size 3-inch to size 16-inch-diameter pipe. The turning device shall be capable of rotating the pipe from .03 to 3 revolutions per minute. The turning device shall be capable of rotating a pipe of not less than 1000 pounds. Constant speed at any speed setting shall be controlled by solid state electronic controls located on the master control. The turning

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device shall control the rotational speed to plus or minus 1 percent of preset values.

3.5.4 Welding heads or torches. The welding heads or torches shall consist of the following:

Type I gas mecal arc (MIG) torch or gas tungsten arc (TIG) torch mounted on a travel device.

Type II gas tungsten arc (TIG) torch enclosed in a unit which revolves around the pipe or tube.

Type III gas metal arc (MIG) or gas tungsten arc torch (TIG) mounted on a torch arm assembly.

The welding head for type II shall be contained and insulated within a split type precision machined housing of high heat and wear resistant material. The electrode holder shall be driven around the pipe or tube by a miniature high torque DC gear motor with tachometer mounted within the welding head handle. The welding head for types II and III shall have a controlled side-to-side motion from near zero to 3/4 inch; controlled oscillation from near zero to 120 cycles per minute; and controlled dwell at each end of the oscillation cycle. Changing of welding heads for type III units shall not require more than 5 minutes. All operations of the welding heads (except manual adjustments) shall be controlled at the master control or pendant control device. The current capacity of the welding head shall be as specified (see 6.2.1).

3.5.5 Torch arm assembly. The welding torch arm assembly for type III pipe welding equipment shall have manual and motorized controls. The master control shall have an override switch for manual control of the automatic welding head (arc length controller) motor in order to vertically adjust the position of the torch prior to and during welding. Override shall be operational over total welding head adjusting range. Adjustment shall be as follows:

a. The torch holder and oscillator assembly shall have a manual or motorized horizontal adjustment range of from not less than 30 inches to not more than 40 inches from the vertical support.

b. The torch holder and oscillator assembly shall have a lateral adjustment positioning range of not less than 4 inches ±2 inch from center. The holder and assembly shall be motorized and controllable from the master control console with a speed range of from not more than 12 inches to not less than 15 inches per minute.

c. The arm assembly shall have a vertical adjustment range of from not more than 20 inches to not less than 54 inches from the floor. The arm assembly shall be motorized and controllable from the master console with a fixed speed of approximately 36 inches per minute. A constant arc voltage shall be maintained with an automatic voltage control welding head (or sensing head) which raises or lowers the torch arm assembly in a vertical plane with a sen-

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sitivity of arc volt variation and correction rate of a range from not less than 12 inches to not more than 15 inches per minute.

d. The vertical column shall be capable of being rotated through a horizontal arc of 180 degrees (90 degrees each side of center) and have the capability of being locked rigidly at any location within this range.

3.5.6 Oscillator. Type III equipment shall have an oscillator with at least a 3/4-inch side-to-side travel and independently controlled dwell at each end of the stroke. The oscillating speed shall be 120 cycles per minute at zero dwell.

3.5.7 Cables and hoses. The welding cables and hoses from supply to torch shall have a length of not less than 25 feet and shall have standard fittings on each end. Torch cables and hoses for type III units shall have sufficient length to allow full extension of torch arm with a minimum of excess.

3.5.8 Wire feeder system. When specified (see 6.2.1), a wire feed system which will feed wire 0.035 inch to 0.062 inch diameter at the minimum rate of from not more than 3 inches to not less than 80 inches per minute shall be provided for the TIG welding. A wire feed unit shall be furnished for all MIG welding equipment. The wire feed unit shall be controlled at the master control console, or from the pendant. The wire feed system shall be provided with a means of adjusting the vertical and lateral movement of filler wire ± 1/16 inch.

3.5.9 Water coolant pump and tank. Water cooled equipment shall have a recirculating water coolant system consisting of a water storage tank, electric motor driven pump, pressure regulator, and strainer or filter. The system shall be of adequate capacity to provide the proper cooling of the welding torch at all times. A control device shall be provided to automatically terminate welding current if coolant pressure or flow rate falls below or above the level necessary to maintain safe operation. The cooling system shall be capable of operation in ambient temperatures from zero to 125 degrees F. All connections and hoses shall be furnished with the equipment.

3.6 Electrical equipment. Unless otherwise specified (see 6.2.1), the electrical requirements shall conform with the requirements of this paragraph. All electrical components including motors starters, relays, switches, and wiring shall conform to and shall be installed in accordance with ANSI/NFPA 79. The equipment shall be completely wired through the controls and ready for connection to the applicable power supply. When the master control and power supply are separate units, the electrical connections shall be made through a disconnect type connection. All electrical characteristics shall be compatible with the power supply specified for the welding equipment.

3.7 Standard equipment. Unless otherwise specified, all standard equipment normally furnished by the manufacturer shall be furnished. Additional equipment required shall be furnished as specified and fully described (see 6.2.1).

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3.8 Marking on plates and charts. All word on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.9 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each welding unit and contain the information listed below. If the welding unit is a special model, the model designation shall include the model of the basic standard welding unit and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be attached in such a way to provide for adjustment to reflect maximum light on the work area. If a built-in light or lighting system is provided to illuminate the work area, the adjustable light described above shall not be required.

3.11 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the welding equipment and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirement of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer' designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Examination. The welding equipment shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests. Except as otherwise specified (see 6.2.1), all tests shall follow the procedures of ANSI NEMA EW-1.

4.7.1 Welding performance tests. The following welding tests shall be performed. The welding equipment shall be placed in position and two consecutive welds shall be made joining applicable diameter mild steel pipe, or applicable diameter copper-nickel tubing. The welds shall be sound and free from defects. The welding tests shall conform to the requirements for P-1 pipe welding as set forth in MIL-STD-278 (SH). The welds shall be made with pipe in the horizontal position. The welded joints shall be radiographed and meet the radiographic acceptance standards for welded P-1 piping. To determine repeatability, only minor alignment adjustments shall be made in the equipment

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when making the two weldments. All applicable components shall be observed to determine compliance with the requirements specified herein.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The pipe or tube welding equipment covered by this specification is intended for use in welding either pipe or tubing in the stationary or revolving positions. Type I and type II equipment is intended for use in welding stationary pipe or tubing where space around the joint is limited Type III equipment is intended for use in welding pipe or tubing that is being rotated by the equipment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type required (see 1.2).

c. First article, if required (see 3.1 and 4.4).

d. If welding equipment is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.2).

e. Additional safety and health requirements, if required (see 3.2.4).

f. Painting, if different (see 3.3.5).

g. Power supply, specify (see 3.5.1).

h. Pendant control, if required (see 3.5.2).

i. Welding head current capacity, specify (see 3.5.4).

j. Wire feeder system, if required (see 3.5.8).

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k. Electrical requirements, if different (see 3.6).

l. Standard equipment, if different (see 3.7).

m. Nameplate, if different (see 3.9).

n. Worklight, if not required (see 3.10).

o. Inspection condition, if different (see 4.3).

p. Quality conformance inspection, if different (see 4.5).

q. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the welding equipment itself. It does not require compliance with those OSHA requirements that concern "the welding equipment in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual welding equipment in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirement that will integrate the new welding equipment into its future operating environment. The welding equipment shall be equipped with all point-of-operation guarding normally furnished as standard equipment supplied to the commercial market. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(e).

6.5 Subject term (key word) listing.

Welding Equipment, Arc, Pipe or Tube
Assembly, Torch Arm
Control, Master
Oscillator
Power Source, Welding
System, Wire Feeder
Device, Travel
Torches, Welding heads

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Army - CE
Navy - YD, AS,OS

Preparing activity:
DLA - IP

Project (3431-0171)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80103C	Welding Equipment, Arc, Pipe or Tube, Automatic

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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